

Smart Electrical Machines



DEFINITION

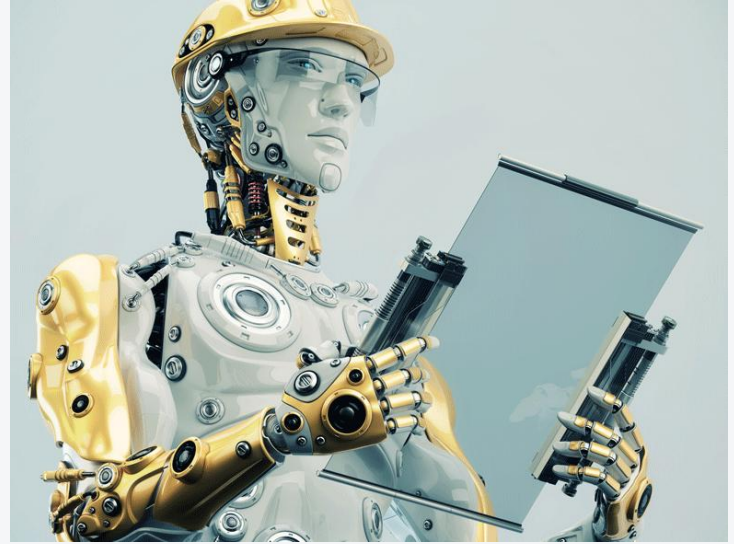
What Are Smart Electrical Machines? Machines?

Traditional Machines

- Fixed-speed operation
- Passive monitoring only
- Reactive maintenance
- No adaptive control

Smart Machines

- Real-time control
- Predictive fault detection
- Adaptive energy optimization
- Self-tuning parameters



THE MERGE

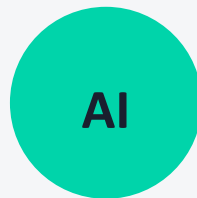
Hardware Meets Artificial Intelligence



Electrical Engineering

Motors, drives, power electronics, magnetic design, and thermal
thermal management

+



Machine Learning

Neural networks, signal processing, optimization algorithms, and
edge inference



MARKET CONTEXT

Why This Major Matters Now

\$870B

Global smart motor market by 2032,
growing at 16% CAGR

70%

Reduction in unplanned downtime with
predictive AI maintenance



LOOKING AHEAD

The Future Potential of This Field

1

Humanoid Robots

2

Digital Twins

3

Quantum-Assisted Design

4

Autonomous Drive Systems

Predictive Maintenance

30-40%

Maintenance cost reduction

2-4 wk

Early failure warning

99.5%

Fault detection accuracy



REAL-WORLD APPLICATION

Energy Optimization

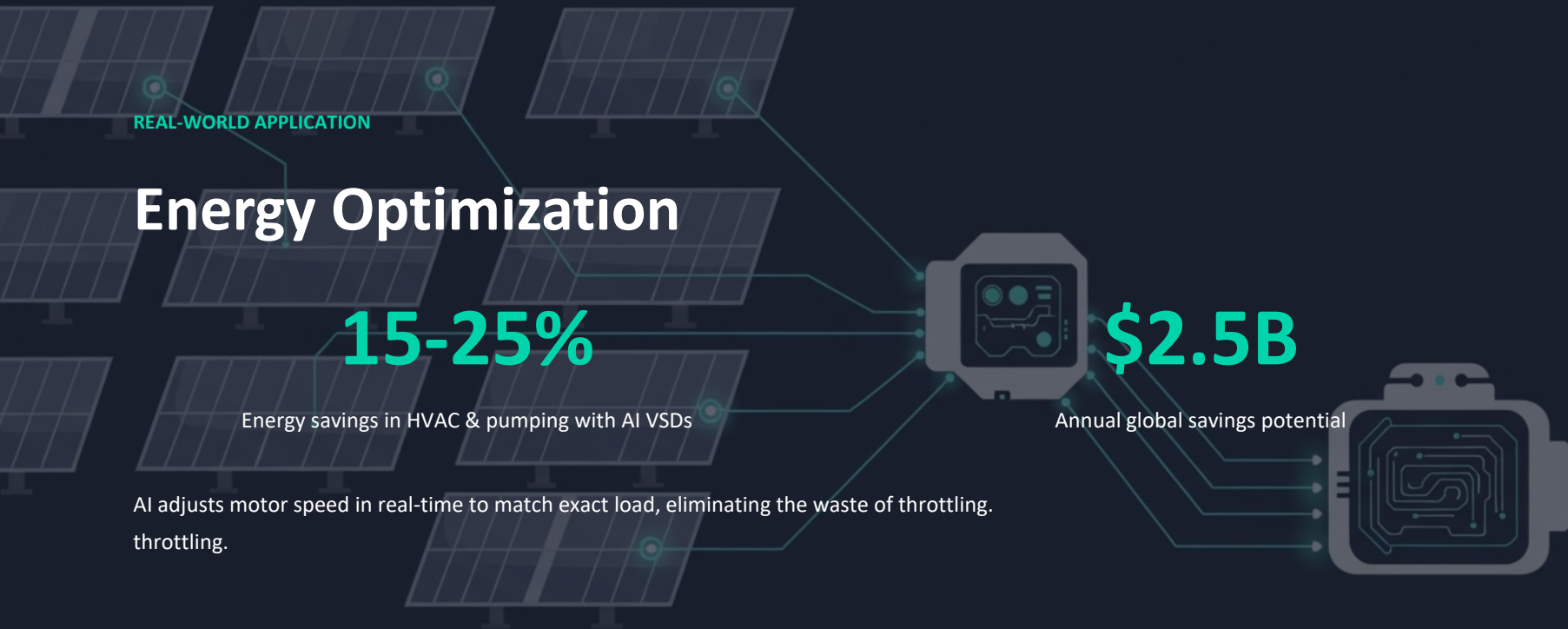
15-25%

Energy savings in HVAC & pumping with AI VSDs

AI adjusts motor speed in real-time to match exact load, eliminating the waste of throttling.
throttling.

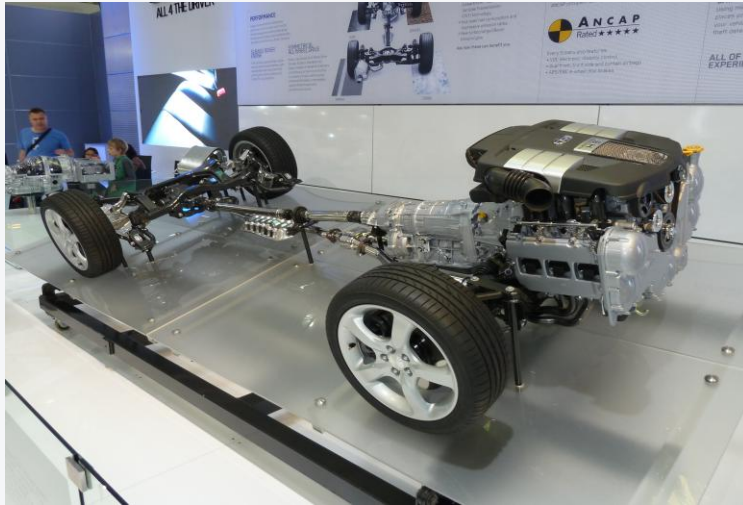
\$2.5B

Annual global savings potential



REAL-WORLD APPLICATION

Electric Vehicle Powertrains



Torque Vectoring

AI distributes torque across wheels independently for maximum traction.

Regenerative Braking AI

Neural networks predict deceleration needs and optimize energy recovery.

Thermal Management

AI-controlled cooling anticipates heat from driving style and road conditions.

REAL-WORLD APPLICATION

Robotics & High-Performance Drives



Cobots

\$8B market by 2030

Surgical Robotics

0.1mm precision

Drone Propulsion

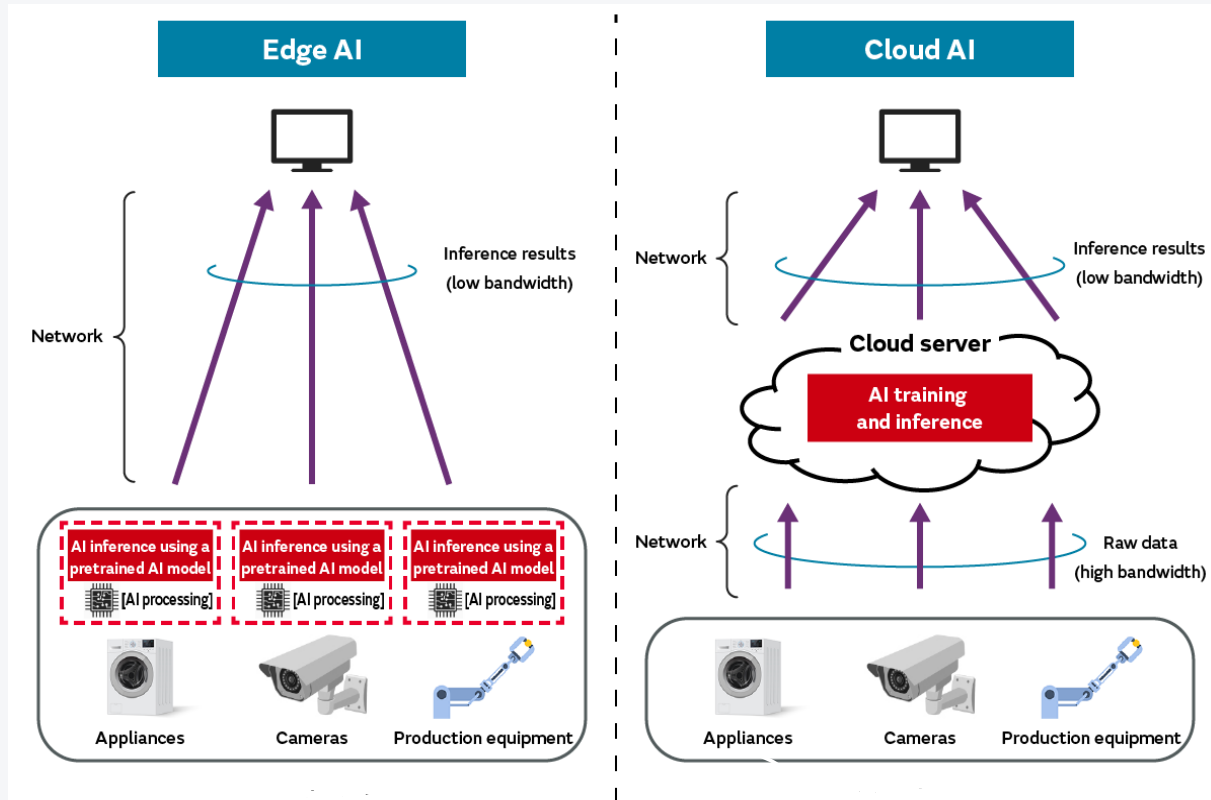
+20-30% flight time

Edge AI in Action

Intelligence that runs on the machine itself, not in the cloud.

Action

What Is Edge AI?



The Edge AI Stack

TinyML Models

TensorFlow Lite, ONNX Micro

Neural Processing Units

ARM Ethos, Cadence Tensilica

Microcontrollers

STM32, ESP32, Raspberry Pi Pico

Sensors & Actuators

Current, vibration, temperature

CASE STUDY

The TinyML Motor Fault Detector

A team at MIT built a motor fault detection system that runs entirely on a \$2 microcontroller, no cloud connection needed.

How It Works

- 1 Hall-effect sensor measures motor current at 10kHz
- 2 FFT extracts frequency signatures from the current waveform
- 3 A quantized neural network (18KB model) classifies fault types
- 4 Decision is made in under 2 milliseconds on-device



18KB

Neural network model size

\$2

Total hardware cost per unit

The skills, projects, and experiences that
that separate good students from
exceptional engineers.

FOUNDATION

Core Skills to Build

E

Electrical

Electric machines theory

Power electronics

Motor control (

Compute

C

Computing

C/C++ & RTOS

Microcontrollers

FPGA basics

Signal processing

AI

Machine Learning

Python, PyTorch

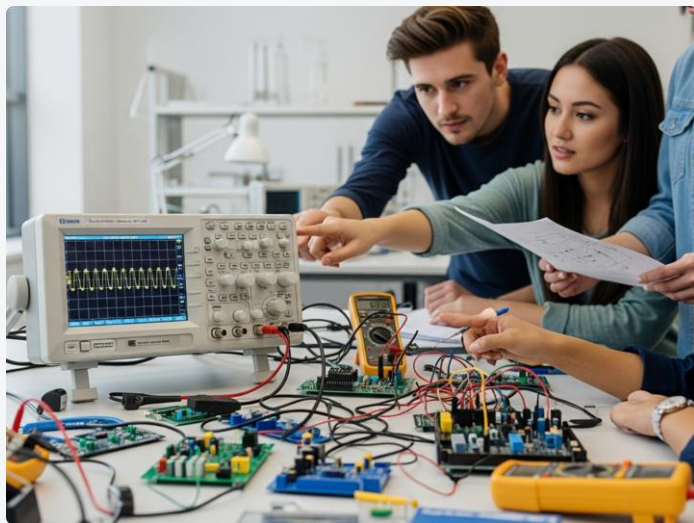
Neural networks

Time-series analysis

Model quantization

PORTFOLIO BUILDING

Projects That Stand Out



BLDC Controller

FOC on STM32

MPPT Converter

Solar optimization

Digital Twin

Simulink + hardware

Fault Classifier

CNN on Pi Pico

Voice Robot Arm

TinyML + servos

Wireless Monitor

ESP32 + LoRaWAN

COMPETITIONS

Competitions to Join

Student Clubs Contests

Future Energy Challenge, Zucker Student Design, and PES competitions for motor drives and power systems.

Formula SAE Electric

Design and race an EV. Motor controller and battery management are core technical challenges.

TinyML Contests

Build the smallest, most efficient ML models for embedded devices. Perfect for motor applications.

Competition wins get you past every HR filter.

CAREER PREPARATION

Internships & Research

Target Companies

ABB, Siemens, Schneider

Tesla, Rivian, BYD

WEG, Nidec, Danfoss

National labs & institutes

Research Areas

Motor design optimization

Fault detection algorithms

Wide bandgap (SiC, GaN)

Model predictive control

Research experience gives a

3x

higher interview rate

DIFFERENTIATION

How to Stand Out

1

Public Portfolio

GitHub with schematics, code, and demo videos.

2

Open Source

Arduino, PlatformIO, TinyML contributions.

3

Teach Publicly

Blog or YouTube explaining motor control.

4

Certifications

MATLAB, PLC, embedded systems.

5

Adjacent Skills

Wireless, power systems, control theory.

6

Network

IEEE conferences, maker spaces, referrals.

ACTION PLAN

Your 4-Year Roadmap

Y1

Foundation

Master circuit theory and C programming

Join IEEE student branch

Build simple Arduino projects

Learn Python basics

Y2

Application

Take electric machines course course

First competition entry

Build a motor controller

Start ML fundamentals

Y3

Integration

Join a research lab

Combine ML with motor control

Summer internship

Publish or present work

Y4

Launch

Capstone: AI motor system

Full-time offer or grad school

Polished portfolio & GitHub

Strong professional network

The Machines Are Getting Smarter.

Will You Keep Up?

Thank You.

msratahmad@gmail.com



Ahmad Masarrat

Satellite Communications Engineer | AI on Edge |
Hardware | R&D | RA

